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<https://swayam.gov.in>https://swayam.gov.in/nc_details/NPTEL

riya.soni17oct@gmail.com ▾

NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction To Machine Learning - IITKGP (course)



Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 0 ()

Week 1 ()

Week 2 ()

Week 3 ()

- ☐ Lecture 12: k-Nearest Neighbour (unit? unit=23&lesson=24)

- ☐ Lecture 13: Feature Selection (unit? unit=23&lesson=25)

Week 3 : Assignment 3

[Ask AI](#)

The due date for submitting this assignment has passed.

Due on 2024-08-14, 23:59 IST.

As per our records you have not submitted this assignment.

1)

2 points

Suppose, you have been given the following data where x_1 and x_2 are the 2 input variables and Class is the dependent variable.

x_1	x_2	Class
-1	1	-
0	1	+
0	2	-
1	-1	-
1	0	+
1	2	+
2	2	-
2	3	+

What will be the class of a new data point $x_1=1$ and $x_2=1$ in 5-NN (k nearest neighbour with $k=5$) using euclidean distance measure?

- A. + Class
- B. - Class
- C. Cannot be determined

☐ A.

☐ Lecture 14:
Feature
Extraction
(unit?
unit=23&lesso
n=26)

☐ Lecture 15:
Collaborative
Filtering (unit?
unit=23&lesso
n=27)

☐ Lecture 16:
Python
Exercise on
kNN and PCA
(unit?
unit=23&lesso
n=28)

☐ Lecture 17:
Tutorial III
(unit?
unit=23&lesso
n=29)

☐ Week 3 :
Lecture
Materials
(unit?
unit=23&lesso
n=30)

☐ Quiz: Week 3
: Assignment
3
(assessment?
name=144)

☐ Feedback
Form for Week
3 (unit?
unit=23&lesso
n=122)

☐ Assignment 3
Solution (unit?
unit=23&lesso
n=154)

Week 4 ()

Week 5 ()

Week 6 ()

Week 7 ()

☐ B.

☐ C.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A.

2)

2 points

Imagine you are dealing with a 10 class classification problem. What is the maximum number of discriminant vectors that can be produced by LDA?

A. 20

B. 14

C. 9

D. 10

☐ A.

☐ B.

☐ C.

☐ D.

No, the answer is incorrect.

Score: 0

Accepted Answers:

C.

3)

Fill in the blanks:

2 points

K-Nearest Neighbor is a _____ , _____ algorithm

A. Non-parametric, eager

B. Parametric, eager

C. Non-parametric, lazy

D. Parametric, lazy

☐ A.

☐ B.

☐ C.

☐ D.

No, the answer is incorrect.

Score: 0

Accepted Answers:

C.

4)

2 points

Week 8 ()

Assignment
Solution ()Download
Videos ()Problem
Solving
Session -
July 2024 ()

Which of the following statements is True about the KNN algorithm?

- A. KNN algorithm does more computation on test time rather than train time.
- B. KNN algorithm does lesser computation on test time rather than train time.
- C. KNN algorithm does an equal amount of computation on test time and train time.
- D. None of these.

- ☐ A.
- ☐ B.
- ☐ C.
- ☐ D.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A.

5)

2 points

Which of the following necessitates feature reduction in machine learning?

- 1. Irrelevant and redundant features
- 2. Curse of dimensionality
- 3. Limited computational resources.

- A. 1 only
- B. 2 only
- C. 1 and 2 only
- D. 1, 2 and 3

- ☐ A.
- ☐ B.
- ☐ C.
- ☐ D.

No, the answer is incorrect.

Score: 0

Accepted Answers:

D.

6)

2 points

When there is noise in data, which of the following options would improve the performance of the k-NN algorithm?

- A. Increase the value of k
- B. Decrease the value of k
- C. Changing value of k will not change the effect of the noise
- D. None of these

- ☐ A.
- ☐ B.
- ☐ C.
- ☐ D.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A.

7)

2 points

Find the value of the Pearson's correlation coefficient of X and Y from the data in the following table.

AGE (X)	GLUCOSE (Y)
43	99
21	65
25	79
42	75

A. 0.47

B. 0.68

C. 1

D. 0.33

☐ A.

☐ B.

☐ C.

☐ D.

No, the answer is incorrect.

Score: 0

Accepted Answers:

B.

8)

2 points

Which of the following statements is/are true about PCA?

1. PCA is a supervised method
2. It identifies the directions that data have the largest variance
3. Maximum number of principal components $\leq$ number of features
4. All principal components are orthogonal to each other

A. Only 2

B. 1, 3 and 4

C. 1, 2 and 3

D. 2, 3 and 4

☐ A.

☐ B.

☐ C.

☐ D.

No, the answer is incorrect.

Score: 0

Accepted Answers:

D.

9)

2 points

In user-based collaborative filtering based recommendation, the items are recommended based on :

- A. Similar users
- B. Similar items
- C. Both of the above
- D. None of the above

☐ A.

☐ B.

☐ C.

☐ D.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A.

10)

2 points

Identify whether the following statement is true or false?

"Linear Discriminant Analysis (LDA) is a supervised method"

- A. TRUE
- B. FALSE

☐ A.

☐ B.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A.

